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Wireshark - Packet 1

No.	Time	Source	Destination	Protocol	Length	Info
3	0.204852	192.168.86.68	224.0.0.251	PIMv6	139	Standard query #0x000 PFR companion-link_tcp.local, "QI" question PFR_sleep-proxy_udp.local, "QI" question OPT
4	0.205172	fe80::17a:1a73:1e7b::f661:f9	:::	PIMv6	159	Standard query #0x000 PFR companion-link_tcp.local, "QI" question PFR_sleep-proxy_udp.local, "QI" question OPT
43	1.024256	0.0.0.0	255.255.255.255	DHCP	286	DHCP Discover > Transaction ID 0x0000c4
44	1.065837	192.168.86.61	128.119.245.12	UDP	78	64928 > 33435 len=28
45	1.066048	192.168.86.1	192.168.86.61	ICMP	98	Time-to-live exceeded (Time to live exceeded in transit)
46	1.069171	192.168.86.61	192.168.86.1	DNS	85	Standard query #0d75d PTR 1.86.168.192.in-addr.arpa
47	1.072918	192.168.86.1	192.168.86.61	DNS	85	Standard query response #0d75d No such name PTR 1.86.168.192.in-addr.arpa
49	1.074616	192.168.86.61	128.119.245.12	UDP	78	64928 > 33435 len=28
49	1.075155	192.168.86.1	192.168.86.61	ICMP	98	Time-to-live exceeded (Time to live exceeded in transit)
50	1.076041	192.168.86.61	128.119.245.12	UDP	78	64928 > 33437 len=28
51	1.076537	192.168.86.1	192.168.86.61	ICMP	98	Time-to-live exceeded (Time to live exceeded in transit)
52	1.076728	192.168.86.61	128.119.245.12	UDP	78	64928 > 33438 len=28
53	1.076909	192.168.86.1	192.168.86.61	ICMP	98	Time-to-live exceeded (Time to live exceeded in transit)
54	1.081113	192.168.86.61	192.168.86.1	DNS	81	Standard query #0d62d PTR 1.0.0.10.in-addr.arpa
55	1.082256	192.168.86.1	192.168.86.61	DNS	81	Standard query response #0d62d No such name PTR 1.0.0.10.in-addr.arpa
56	1.082567	192.168.86.61	128.119.245.12	UDP	78	64928 > 33439 len=28
57	1.083800	192.168.86.1	192.168.86.61	ICMP	98	Time-to-live exceeded (Time to live exceeded in transit)
58	1.089002	192.168.86.61	128.119.245.12	UDP	78	64928 > 33440 len=28
59	1.090000	192.168.86.1	192.168.86.61	ICMP	98	Time-to-live exceeded (Time to live exceeded in transit)
60	1.092056	192.168.86.61	128.119.245.12	UDP	78	64928 > 33441 len=28
61	1.093167	96.128.68.9	192.168.86.61	ICMP	78	Time-to-live exceeded (Time to live exceeded in transit)
62	1.093656	192.168.86.61	128.119.245.12	UDP	78	64928 > 33442 len=28
63	1.097290	96.128.68.9	192.168.86.61	ICMP	78	Time-to-live exceeded (Time to live exceeded in transit)
64	1.028173	192.168.86.61	128.119.245.12	UDP	78	64928 > 33443 len=28
65	1.098130	96.128.68.9	192.168.86.61	ICMP	78	Time-to-live exceeded (Time to live exceeded in transit)
66	1.098320	96.128.68.9	128.119.245.12	ICMP	78	Time-to-live exceeded (Time to live exceeded in transit)

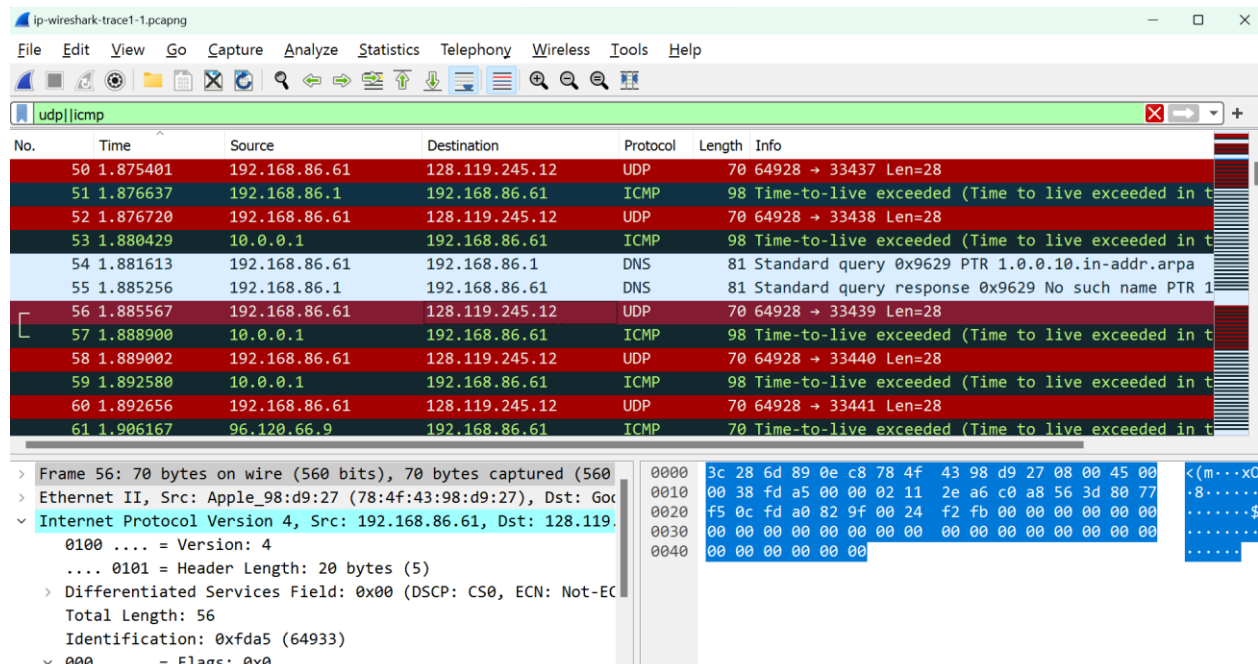
Frame 44: 78 bytes on wire (568 bits), 70 bytes captured (568 bits) on Interface em0, id 0
Ethernet II, Src: Apple_Mac@9:27:78:4d:43:8b, Dst: Google_Bus@dc:c3:28:6d:8bdc5d

```

Internet Protocol Version 4, Src: 192.168.86.61, Dst: 128.119.245.12
0180 .... : Version: 4
      0100 ...: Header Length: 20 bytes (5)
> Differentiated Services Field: 0xb0 (DSCP: CS0, ECN: Not-ECT)
Total Length: 56
Identification: bufaf1 (44020)
0000 .... : Flags: 0xb0
      0... ..: Reserved bit: Not set
      .0....: Don't Fragment: Not set
      .0....: More Fragments: Not set
      ...000000000000 : Fragment Offset: 0
> Time to Live: 1
    Protocol: UDP (17)
    Header Checksum: 0x2fa [validation disabled]
    [Header checksum status: Unverified]
    Source Address: 192.168.86.61
    Destination Address: 128.119.245.12
> User Datagram Protocol, Src Port: 64928, Dst Port: 33435
> Data (28 bytes)

```

Packet details show a series of ICMP Echo requests from 96.128.68.9 to 192.168.86.61, which are being blocked by a firewall rule (likely blocking traffic to port 33435). The packet capture shows the source IP changing between frames, suggesting multiple hosts or a spoofed source.



1. Select the first UDP segment sent by your computer via the `traceroute` command to `gaia.cs.umass.edu`. (Hint: this is 44th packet in the trace file in the `ipwireshark-trace1-1.pcapng` file). Expand the Internet Protocol part of the packet in the packet details window. What is the IP address of your computer?

ANSWER:

The IP of my computer is 192.168.86.61[IPV4].

2. What is the value in the time-to-live (TTL) field in this IPv4 datagram's header?

ANSWER:

TTL: 4

3. What is the value in the upper layer protocol field in this IPv4 datagram's header?

ANSWER:

Upper layer protocol field in IPV4 datagram's header: UDP(17).

4. How many bytes are in the IP header?

ANSWER:

IP Header length: 20bytes.

5. How many bytes are in the payload of the IP datagram? Explain how you determined the number of payload bytes.

ANSWER:

Total Length = 56

Data Length = Total length – Header length

= 56-20

= 36

6. Has this IP datagram been fragmented? Explain how you determined whether or not the datagram has been fragmented.

ANSWER:

Since more fragment and don't fragment are not set, the datagram has not been fragmented.

The image shows a Wireshark packet capture window titled "ip-wireshark-trace-1-1.pcapng". The packet list on the left shows a series of UDP packets from source 192.168.86.61 to destination 128.119.245.12. Packet 52 is selected, showing details for Ethernet II, Internet Protocol Version 4, and User Datagram Protocol. The IP header shows a total length of 56, identification of 0xfda4, and flags of 0x0. The UDP header shows source port 64928 and destination port 33438. The packet bytes pane on the right shows the raw data in hexadecimal and ASCII.

No.	Time	Source	Destination	Protocol	Length	Info
44	1.865637	192.168.86.61	128.119.245.12	UDP	70	64928 → 33435 Len=28
48	1.871615	192.168.86.61	128.119.245.12	UDP	70	64928 → 33436 Len=28
50	1.875491	192.168.86.61	128.119.245.12	UDP	70	64928 → 33437 Len=28
52	1.876720	192.168.86.61	128.119.245.12	UDP	70	64928 → 33438 Len=28
56	1.885567	192.168.86.61	128.119.245.12	UDP	70	64928 → 33439 Len=28
58	1.889802	192.168.86.61	128.119.245.12	UDP	70	64928 → 33440 Len=28
60	1.892656	192.168.86.61	128.119.245.12	UDP	70	64928 → 33441 Len=28
62	1.907036	192.168.86.61	128.119.245.12	UDP	70	64928 → 33442 Len=28
64	1.920173	192.168.86.61	128.119.245.12	UDP	70	64928 → 33443 Len=28
67	1.940279	192.168.86.61	128.119.245.12	UDP	70	64928 → 33444 Len=28
69	1.951481	192.168.86.61	128.119.245.12	UDP	70	64928 → 33445 Len=28
71	1.965335	192.168.86.61	128.119.245.12	UDP	70	64928 → 33446 Len=28
73	1.975799	192.168.86.61	128.119.245.12	UDP	70	64928 → 33447 Len=28
75	1.991739	192.168.86.61	128.119.245.12	UDP	70	64928 → 33448 Len=28
77	2.008887	192.168.86.61	128.119.245.12	UDP	70	64928 → 33449 Len=28
79	2.025022	192.168.86.61	128.119.245.12	UDP	70	64928 → 33450 Len=28
81	2.045745	192.168.86.61	128.119.245.12	UDP	70	64928 → 33451 Len=28
83	2.063183	192.168.86.61	128.119.245.12	UDP	70	64928 → 33452 Len=28
85	2.081361	192.168.86.61	128.119.245.12	UDP	70	64928 → 33453 Len=28
87	2.104137	192.168.86.61	128.119.245.12	UDP	70	64928 → 33454 Len=28
89	2.122238	192.168.86.61	128.119.245.12	UDP	70	64928 → 33455 Len=28
91	2.141245	192.168.86.61	128.119.245.12	UDP	70	64928 → 33456 Len=28
93	2.161747	192.168.86.61	128.119.245.12	UDP	70	64928 → 33457 Len=28
95	2.180025	192.168.86.61	128.119.245.12	UDP	70	64928 → 33458 Len=28
97	2.199389	192.168.86.61	128.119.245.12	UDP	70	64928 → 33459 Len=28
100	3.258383	192.168.86.61	128.119.245.12	UDP	70	64928 → 33460 Len=28

Frame 52: 70 bytes on wire (560 bits), 70 bytes captured (560 bits) on interface enb, id 0
> Ethernet II, Src: Apple_98:d9:d9:27 (78:4f:43:98:d9:27), Dst: Google_89:0e:c8 (3c:28:6d:89:0e:c8)
▼ Internet Protocol Version 4, Src: 192.168.86.61, Dst: 128.119.245.12
0100 = Version: 4
.... 0101 = Header Length: 20 bytes (5)
> Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
Total Length: 56
Identification: 0xfda4 (64932)
> 0000 = Flags: 0x0
...0 0000 0000 0000 = Fragment Offset: 0
> Time to Live: 2
Protocol: UDP (17)
Header Checksum: 0x2ea7 [validation disabled]
[Header checksum status: Unverified]
Source Address: 192.168.86.61
Destination Address: 128.119.245.12
> User Datagram Protocol, Src Port: 64928, Dst Port: 33438
> Data (28 bytes)

0000 3c 28 6d 89 0e c8 78 4f 43 98 d9 27 08 00 45 00 <(.....C.....E..
0010 00 38 fd a4 00 00 02 11 2e a7 c0 a8 56 3d 00 77 B.....Vw
0020 f5 0c fd a0 02 9e 00 24 f2 fc 00 00 00 00 00 00\$
0030 00 00 00 00 00 00 00 00 00 00 00 00 00 00
0040 00 00 00 00 00 00 00 00 00 00 00 00 00 00

7. Which fields in the IP datagram *always* change from one datagram to the next within this series of UDP segments sent by your computer destined to 128.119.245.12, via traceroute? Why? (List ALL the fields that change)

ANSWER:

Fields in IP datagram always change from one datagram to the next datagram within series of UDP segments:

- Time To Live:
In received datagram, router decrements TTL by 1.
- Identification:
It should be unique because it is required for reassembly.
- Header Checksum:
It changes since it is reverified & recomputed at each address.

8. Which fields in this sequence of IP datagrams (containing UDP segments) stay constant? Why? (List ALL the fields that stay constant)

Answer:

Fields in IP datagram stay constant:

- Version: Since, here using IPV4
- Header length: since all packets are UDP
- Differentiated Service: Since, all packets are UDP
- Protocol: Since all packets are UDP
- Source address: Since, all packets are sent from my computer
- Destination address: Since, all packets are sent to same host
- Flags : The Don't fragment and more fragment bits are not set because the packets are not fragmented here(as attached in screenshots below).

9. Describe the pattern you see in the values in the Identification field of the IP datagrams being sent by your computer.

Answer:

The screenshot shows a Wireshark capture of network traffic. The top pane displays a list of captured packets. The bottom pane shows the details of a selected packet (Frame 44), which is an Ethernet II frame containing an Internet Protocol Version 4 (IPv4) packet. The packet details are as follows:

- Frame 44: 70 bytes on wire (560 bits), 70 bytes captured (560 bits) on interface en0, id 0
- Ethernet II, Src: Apple_98:d9:27 (78:4f:43:98:d9:27), Dst: Google_89:0e:c8 (3c:28:6d:89:0e:c8)
- Internet Protocol Version 4, Src: 192.168.86.61, Dst: 128.119.245.12
 - 0100 = Version: 4
 - 0101 = Header Length: 20 bytes (5)
 - > Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
 - Total Length: 56
 - Identification: 0x6fa1 (64929)
 - 0000 = Flags: 0x0
 - 0... = Reserved bit: Not set
 - .0... = Don't fragment: Not set
 - ..0. = More fragments: Not set
 - ...0 0000 0000 0000 = Fragment Offset: 0
 - > Time to Live: 1
 - Protocol: UDP (17)
 - Header Checksum: 0x2faa [validation disabled]
 - [Header checksum status: Unverified]
 - Source Address: 192.168.86.61
 - Destination Address: 128.119.245.12
- User Datagram Protocol, Src Port: 64928, Dst Port: 33435
- Data (28 bytes)

The packet details pane also shows the raw data in hexadecimal and ASCII format.

Wireshark interface showing packet capture data. The top pane displays a list of captured packets, including Ethernet II, Internet Protocol Version 4, and User Datagram Protocol (UDP). The middle pane shows the details of the selected packet (Frame 48), including Ethernet II, Internet Protocol Version 4, and User Datagram Protocol (UDP). The bottom pane displays the raw packet data in hexadecimal and ASCII.

No.	Time	Source	Destination	Protocol	Length	Info
44	1.865637	192.168.86.61	128.119.245.12	UDP	70	64928 → 33435 Len=28
48	1.874016	192.168.86.61	128.119.245.12	UDP	70	64928 → 33436 Len=28
50	1.875401	192.168.86.61	128.119.245.12	UDP	70	64928 → 33437 Len=28
52	1.876728	192.168.86.61	128.119.245.12	UDP	70	64928 → 33438 Len=28
56	1.885567	192.168.86.61	128.119.245.12	UDP	70	64928 → 33439 Len=28
58	1.885902	192.168.86.61	128.119.245.12	UDP	70	64928 → 33440 Len=28
60	1.892656	192.168.86.61	128.119.245.12	UDP	70	64928 → 33441 Len=28
62	1.907036	192.168.86.61	128.119.245.12	UDP	70	64928 → 33442 Len=28
64	1.920173	192.168.86.61	128.119.245.12	UDP	70	64928 → 33443 Len=28
67	1.940279	192.168.86.61	128.119.245.12	UDP	70	64928 → 33444 Len=28
69	1.951481	192.168.86.61	128.119.245.12	UDP	70	64928 → 33445 Len=28
71	1.965335	192.168.86.61	128.119.245.12	UDP	70	64928 → 33446 Len=28
73	1.975799	192.168.86.61	128.119.245.12	UDP	70	64928 → 33447 Len=28
75	1.991739	192.168.86.61	128.119.245.12	UDP	70	64928 → 33448 Len=28
77	2.000887	192.168.86.61	128.119.245.12	UDP	70	64928 → 33449 Len=28
79	2.025022	192.168.86.61	128.119.245.12	UDP	70	64928 → 33450 Len=28
81	2.045745	192.168.86.61	128.119.245.12	UDP	70	64928 → 33451 Len=28
83	2.063103	192.168.86.61	128.119.245.12	UDP	70	64928 → 33452 Len=28
85	2.081361	192.168.86.61	128.119.245.12	UDP	70	64928 → 33453 Len=28
87	2.104137	192.168.86.61	128.119.245.12	UDP	70	64928 → 33454 Len=28
89	2.122238	192.168.86.61	128.119.245.12	UDP	70	64928 → 33455 Len=28
91	2.141245	192.168.86.61	128.119.245.12	UDP	70	64928 → 33456 Len=28
93	2.161747	192.168.86.61	128.119.245.12	UDP	70	64928 → 33457 Len=28
95	2.180825	192.168.86.61	128.119.245.12	UDP	70	64928 → 33458 Len=28
97	2.199309	192.168.86.61	128.119.245.12	UDP	70	64928 → 33459 Len=28

Frame 48: 70 bytes on wire (560 bits), 70 bytes captured (560 bits) on interface em0, id 0
Ethernet II, Src: Apple_90:d9:d7 (78:4f:43:98:d9:d7), Dst: Google_89:0e:c8 (3c:28:dd:89:0e:c8)
Internet Protocol Version 4, Src: 192.168.86.61, Dst: 128.119.245.12
..... = Version: 4
..... = Header Length: 20 bytes (5)
..... = Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
Total Length: 56
Identification: 0xfda2 (64930)
..... = Flags: 0x0
..... = Reserved bit: Not set
..... = Don't fragment: Not set
..... = More fragments: Not set
..... = Fragment Offset: 0
Time to Live: 1
Protocol: UDP (17)
Header Checksum: 0x2fa0 [validation disabled]
[Header checksum status: Unverified]
Source Address: 192.168.86.61
Destination Address: 128.119.245.12
User Datagram Protocol, Src Port: 64928, Dst Port: 33436
Data (28 bytes)

Wireshark interface showing packet capture data. The top pane displays a list of captured packets, including Ethernet II, Internet Protocol Version 4, and User Datagram Protocol (UDP). The middle pane shows the details of the selected packet (Frame 50), including Ethernet II, Internet Protocol Version 4, and User Datagram Protocol (UDP). The bottom pane displays the raw packet data in hexadecimal and ASCII.

No.	Time	Source	Destination	Protocol	Length	Info
44	1.865637	192.168.86.61	128.119.245.12	UDP	70	64928 → 33435 Len=28
48	1.874016	192.168.86.61	128.119.245.12	UDP	70	64928 → 33436 Len=28
50	1.875401	192.168.86.61	128.119.245.12	UDP	70	64928 → 33437 Len=28
52	1.876728	192.168.86.61	128.119.245.12	UDP	70	64928 → 33438 Len=28
56	1.885567	192.168.86.61	128.119.245.12	UDP	70	64928 → 33439 Len=28
58	1.885902	192.168.86.61	128.119.245.12	UDP	70	64928 → 33440 Len=28
60	1.892656	192.168.86.61	128.119.245.12	UDP	70	64928 → 33441 Len=28
62	1.907036	192.168.86.61	128.119.245.12	UDP	70	64928 → 33442 Len=28
64	1.920173	192.168.86.61	128.119.245.12	UDP	70	64928 → 33443 Len=28
67	1.940279	192.168.86.61	128.119.245.12	UDP	70	64928 → 33444 Len=28
69	1.951481	192.168.86.61	128.119.245.12	UDP	70	64928 → 33445 Len=28
71	1.965335	192.168.86.61	128.119.245.12	UDP	70	64928 → 33446 Len=28
73	1.975799	192.168.86.61	128.119.245.12	UDP	70	64928 → 33447 Len=28
75	1.991739	192.168.86.61	128.119.245.12	UDP	70	64928 → 33448 Len=28
77	2.000887	192.168.86.61	128.119.245.12	UDP	70	64928 → 33449 Len=28
79	2.025022	192.168.86.61	128.119.245.12	UDP	70	64928 → 33450 Len=28
81	2.045745	192.168.86.61	128.119.245.12	UDP	70	64928 → 33451 Len=28
83	2.063103	192.168.86.61	128.119.245.12	UDP	70	64928 → 33452 Len=28
85	2.081361	192.168.86.61	128.119.245.12	UDP	70	64928 → 33453 Len=28
87	2.104137	192.168.86.61	128.119.245.12	UDP	70	64928 → 33454 Len=28
89	2.122238	192.168.86.61	128.119.245.12	UDP	70	64928 → 33455 Len=28
91	2.141245	192.168.86.61	128.119.245.12	UDP	70	64928 → 33456 Len=28
93	2.161747	192.168.86.61	128.119.245.12	UDP	70	64928 → 33457 Len=28
95	2.180825	192.168.86.61	128.119.245.12	UDP	70	64928 → 33458 Len=28
97	2.199309	192.168.86.61	128.119.245.12	UDP	70	64928 → 33459 Len=28

Frame 50: 70 bytes on wire (560 bits), 70 bytes captured (560 bits) on interface em0, id 0
Ethernet II, Src: Apple_90:d9:d7 (78:4f:43:98:d9:d7), Dst: Google_89:0e:c8 (3c:28:dd:89:0e:c8)
Internet Protocol Version 4, Src: 192.168.86.61, Dst: 128.119.245.12
..... = Version: 4
..... = Header Length: 20 bytes (5)
..... = Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
Total Length: 56
Identification: 0xfda3 (64931)
..... = Flags: 0x0
..... = Reserved bit: Not set
..... = Don't fragment: Not set
..... = More fragments: Not set
..... = Fragment Offset: 0
Time to Live: 1
Protocol: UDP (17)
Header Checksum: 0x2fa0 [validation disabled]
[Header checksum status: Unverified]
Source Address: 192.168.86.61
Destination Address: 128.119.245.12
User Datagram Protocol, Src Port: 64928, Dst Port: 33437
Data (28 bytes)

Wireshark interface showing a packet capture on the 'p-wirehark-trace1-1.pcapng' file. The filter is set to 'ip.src==192.168.86.61 and ip.dst==128.119.245.12 and udp and icmp'. The packet list shows a series of UDP packets from 192.168.86.61 to 128.119.245.12, all with length 28. The selected packet (No. 52) is a UDP packet with source 192.168.86.61 and destination 128.119.245.12, length 28. The packet details pane shows the following structure:

- Frame 52: 70 bytes on wire (560 bits), 70 bytes captured (560 bits) on interface en0, id 0
- Ethernet II, Src: Apple_98:d9:27 (78:4f:43:98:d9:27), Dst: Google_89:0e:c8 (3c:20:6d:89:0e:c8)
- Internet Protocol Version 4, Src: 192.168.86.61, Dst: 128.119.245.12
 - 0000 = Version: 4
 - 0101 = Header Length: 20 bytes (5)
 - Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
 - Total Length: 56
 - Identification: 0xfda4 (64932)
 - 0000 = Flags: 0x0
 - 0... = Reserved bit: Not set
 - .0... = Don't fragment: Not set
 - ..0... = More fragments: Not set
 - ...0 0000 0000 0000 = Fragment Offset: 0
 - Time to Live: 2
 - Protocol: UDP (17)
 - Header Checksum: 0x2ea7 [validation disabled]
 - [Header checksum status: Unverified]
 - Source Address: 192.168.86.61
 - Destination Address: 128.119.245.12
- User Datagram Protocol, Src Port: 64928, Dst Port: 33438
- Data (28 bytes)

The packet bytes pane shows the raw data in hexadecimal and ASCII:

```
0000 3c 20 6d 89 0e c8 78 4f 43 98 d9 27 00 00 45 00 <[...x0 C...:E:
0010 00 38 fd a4 00 00 02 11 2e a7 c0 a8 56 3d 80 77 0.....Vw
0020 f5 0c fd a0 82 9e 00 24 f2 fc 00 00 00 00 00 00 .....$
0030 00 00 00 00 00 00 00 00 00 00 00 00 00 00 .....
0040 00 00 00 00 00 00 00 00 00 00 00 00 00 00 .....
```

The status bar at the bottom indicates 317 packets displayed (73.0%) and the profile is set to Default. The system clock shows 2:27 PM on 11/9/2023.

ip-wireshark-trace1-1.pcapng

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

ip.src==192.168.86.61 and ip.dst==128.119.245.12 and udp and icmp

No.	Time	Source	Destination	Protocol	Length	Info
44	1.865637	192.168.86.61	128.119.245.12	UDP	70	64928 → 33435 Len=28
48	1.874816	192.168.86.61	128.119.245.12	UDP	70	64928 → 33436 Len=28
50	1.875481	192.168.86.61	128.119.245.12	UDP	70	64928 → 33437 Len=28
52	1.876728	192.168.86.61	128.119.245.12	UDP	70	64928 → 33438 Len=28
56	1.885567	192.168.86.61	128.119.245.12	UDP	70	64928 → 33439 Len=28
58	1.889802	192.168.86.61	128.119.245.12	UDP	70	64928 → 33440 Len=28
60	1.892656	192.168.86.61	128.119.245.12	UDP	70	64928 → 33441 Len=28
62	1.907836	192.168.86.61	128.119.245.12	UDP	70	64928 → 33442 Len=28
64	1.928173	192.168.86.61	128.119.245.12	UDP	70	64928 → 33443 Len=28
67	1.940279	192.168.86.61	128.119.245.12	UDP	70	64928 → 33444 Len=28
69	1.951481	192.168.86.61	128.119.245.12	UDP	70	64928 → 33445 Len=28
71	1.965335	192.168.86.61	128.119.245.12	UDP	70	64928 → 33446 Len=28
73	1.975799	192.168.86.61	128.119.245.12	UDP	70	64928 → 33447 Len=28
75	1.991739	192.168.86.61	128.119.245.12	UDP	70	64928 → 33448 Len=28
77	2.008087	192.168.86.61	128.119.245.12	UDP	70	64928 → 33449 Len=28
79	2.025822	192.168.86.61	128.119.245.12	UDP	70	64928 → 33450 Len=28
81	2.045745	192.168.86.61	128.119.245.12	UDP	70	64928 → 33451 Len=28
83	2.065105	192.168.86.61	128.119.245.12	UDP	70	64928 → 33452 Len=28
85	2.082361	192.168.86.61	128.119.245.12	UDP	70	64928 → 33453 Len=28
87	2.104137	192.168.86.61	128.119.245.12	UDP	70	64928 → 33454 Len=28
89	2.122238	192.168.86.61	128.119.245.12	UDP	70	64928 → 33455 Len=28
91	2.141245	192.168.86.61	128.119.245.12	UDP	70	64928 → 33456 Len=28
93	2.161747	192.168.86.61	128.119.245.12	UDP	70	64928 → 33457 Len=28
95	2.180825	192.168.86.61	128.119.245.12	UDP	70	64928 → 33458 Len=28
97	2.199389	192.168.86.61	128.119.245.12	UDP	70	64928 → 33459 Len=28
100	3.550765	192.168.86.61	128.119.245.12	UDP	70	64928 → 33460 Len=28

> Frame 56: 70 bytes on wire (560 bits), 70 bytes captured (560 bits) on interface em0, id 0

> Ethernet II, Src: Apple_98:09:27 (78:4f:43:98:09:27), Dst: Google_09:0e:c8 (3c:28:6d:09:0e:c8)

> Internet Protocol Version 4, Src: 192.168.86.61, Dst: 128.119.245.12

0000 = Version: 4

..... 0001 = Header Length: 20 bytes (5)

> Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)

Total Length: 56

Identification: 0xfda5 (64933)

0000 = Flags: 0x0

0... .. = Reserved bit: Not set

..0... .. = Don't fragment: Not set

...0... .. = More fragments: Not set

...0 0000 0000 0000 = Fragment Offset: 0

> Time to Live: 2

Protocol: UDP (17)

Header Checksum: 0x2ea6 [validation disabled]

[Header checksum status: Unverified]

Source Address: 192.168.86.61

Destination Address: 128.119.245.12

> User Datagram Protocol, Src Port: 64928, Dst Port: 33439

> Data (28 bytes)

0000 3c 28 6d 09 0e c8 78 4f 43 98 09 27 00 00 45 00 <[...>C...E

0010 00 30 fd a5 00 00 02 11 2e a5 c8 a5 56 3d 00 77 8U-W

0020 f5 0c fd a0 82 9f 00 24 f2 fd 00 00 00 00 00 00\$

0030 00 00 00 00 00 00 00 00 00 00 00 00 00 00

0040 00 00 00 00 00 00 00 00 00 00 00 00 00 00

Packets: 317 · Displayed: 73 (23.0%)

Profile: Default

2:27 PM 11/9/2023

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ip.src==192.168.86.61 and ip.dst==128.119.245.12 and udp and icmp

No.	Time	Source	Destination	Protocol	Length	Info
44	1.865637	192.168.86.61	128.119.245.12	UDP	70	64928 → 33435 Len=28
48	1.874816	192.168.86.61	128.119.245.12	UDP	70	64928 → 33436 Len=28
50	1.875481	192.168.86.61	128.119.245.12	UDP	70	64928 → 33437 Len=28
52	1.876728	192.168.86.61	128.119.245.12	UDP	70	64928 → 33438 Len=28
56	1.885567	192.168.86.61	128.119.245.12	UDP	70	64928 → 33439 Len=28
58	1.889802	192.168.86.61	128.119.245.12	UDP	70	64928 → 33440 Len=28
60	1.892656	192.168.86.61	128.119.245.12	UDP	70	64928 → 33441 Len=28
62	1.907836	192.168.86.61	128.119.245.12	UDP	70	64928 → 33442 Len=28
64	1.928173	192.168.86.61	128.119.245.12	UDP	70	64928 → 33443 Len=28
67	1.940279	192.168.86.61	128.119.245.12	UDP	70	64928 → 33444 Len=28
69	1.951481	192.168.86.61	128.119.245.12	UDP	70	64928 → 33445 Len=28
71	1.965335	192.168.86.61	128.119.245.12	UDP	70	64928 → 33446 Len=28
73	1.975799	192.168.86.61	128.119.245.12	UDP	70	64928 → 33447 Len=28
75	1.991739	192.168.86.61	128.119.245.12	UDP	70	64928 → 33448 Len=28
77	2.008087	192.168.86.61	128.119.245.12	UDP	70	64928 → 33449 Len=28
79	2.025822	192.168.86.61	128.119.245.12	UDP	70	64928 → 33450 Len=28

> Frame 64: 70 bytes on wire (560 bits), 70 bytes captured (560 bits) on interface em0, id 0

> Ethernet II, Src: Apple_98:09:27 (78:4f:43:98:09:27), Dst: Google_09:0e:c8 (3c:28:6d:09:0e:c8)

> Internet Protocol Version 4, Src: 192.168.86.61, Dst: 128.119.245.12

0000 = Version: 4

..... 0001 = Header Length: 20 bytes (5)

> Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)

Total Length: 56

Identification: 0xfda9 (64937)

0000 = Flags: 0x0

0... .. = Reserved bit: Not set

..0... .. = Don't fragment: Not set

...0... .. = More fragments: Not set

...0 0000 0000 0000 = Fragment Offset: 0

> Time to Live: 3

Protocol: UDP (17)

Header Checksum: 0xd2a2 [validation disabled]

[Header checksum status: Unverified]

Source Address: 192.168.86.61

Destination Address: 128.119.245.12

> User Datagram Protocol, Src Port: 64928, Dst Port: 33443

> Data (28 bytes)

0000 3c 28 6d 09 0e c8 78 4f 43 98 09 27 00 00 45 00 <[...>C...E

0010 00 30 fd a9 00 00 03 11 2e a2 c8 a5 56 3d 00 77 8U-W

0020 f5 0c fd a0 82 a3 00 24 f2 f7 00 00 00 00 00 00\$

0030 00 00 00 00 00 00 00 00 00 00 00 00 00 00

0040 00 00 00 00 00 00 00 00 00 00 00 00 00 00

No.	Time	Source	Destination	Protocol	Length	Info
44	1.865337	192.168.86.61	128.119.245.12	UDP	70	64928 → 33435 Len=28
48	1.874816	192.168.86.61	128.119.245.12	UDP	70	64928 → 33436 Len=28
50	1.875401	192.168.86.61	128.119.245.12	UDP	70	64928 → 33437 Len=28
52	1.876720	192.168.86.61	128.119.245.12	UDP	70	64928 → 33438 Len=28
56	1.885567	192.168.86.61	128.119.245.12	UDP	70	64928 → 33439 Len=28
58	1.889802	192.168.86.61	128.119.245.12	UDP	70	64928 → 33440 Len=28
60	1.892856	192.168.86.61	128.119.245.12	UDP	70	64928 → 33441 Len=28
62	1.907836	192.168.86.61	128.119.245.12	UDP	70	64928 → 33442 Len=28
64	1.928173	192.168.86.61	128.119.245.12	UDP	70	64928 → 33443 Len=28
67	1.948279	192.168.86.61	128.119.245.12	UDP	70	64928 → 33444 Len=28
69	1.951461	192.168.86.61	128.119.245.12	UDP	70	64928 → 33445 Len=28
71	1.963395	192.168.86.61	128.119.245.12	UDP	70	64928 → 33446 Len=28
73	1.975799	192.168.86.61	128.119.245.12	UDP	70	64928 → 33447 Len=28
75	1.991739	192.168.86.61	128.119.245.12	UDP	70	64928 → 33448 Len=28
77	2.008887	192.168.86.61	128.119.245.12	UDP	70	64928 → 33449 Len=28
79	2.025922	192.168.86.61	128.119.245.12	UDP	70	64928 → 33450 Len=28

The pattern in the values in the Identification field of the IP datagrams being sent by my computer changes in Hexadecimal format (0 to f) and increases by 1 after each packet.

10. What is the upper layer protocol specified in the IP datagrams returned from the routers?

ANSWER:

Upper Layer Protocol: ICMP(1)

11. Are the values in the Identification fields (across the sequence of all of ICMP packets from all of the routers) similar in behavior to your answer to question 9 above?

ANSWER:

No, the values in the identification fields are not similar to question 9.

12. Are the values of the TTL fields similar, across all of ICMP packets from all of the routers?

ANSWER:

No, the values of TTL fields are not similar across all of ICMP packets from all of the routers. It is similar only for same 3 packets as screenshots attached below, but changes after every 3 packets. So, the values of TTL are not similar from all of the routers.

The screenshot displays the Wireshark network protocol analyzer interface. At the top, the title bar reads "ip-wirehark-1.10.4 pcapng". The menu bar includes File, Edit, View, Go, Capture, Analyze, Statistics, Telephony, Wireless, Tools, and Help. Below the menu is a toolbar with various icons for file operations, capture control, and analysis. The main window is divided into three panes:

- Packet List:** Shows a single packet (No. 49) at Time 1.875315, Source 192.168.86.61, Destination 192.168.86.61, Protocol ICMP, Length 96. The Info column indicates "Time-to-live exceeded (Time to live exceeded in transit)".
- Packet Details:** Expands the selected packet, showing:
 - Frame 49: 96 bytes on wire (784 bits), 96 bytes captured (784 bytes) on interface em0, at 0
 - Ethernet II, Src: Qualcomm (96:12:8b:09:00:01), Dst: Apple (08:00:27:7b:14:43:98:09:27)
 - Internet Protocol Version 4, Src: 192.168.86.1, Dst: 192.168.86.61
 - 0800 = Version: 4
 - 0101 = Header Length: 20 bytes (5)
 - > Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
 - Total Length: 94
 - Identification: 0x0000 (26762)
 - 0800 = Flags: 0x0
 - 0... = Reserved bit: Not set
 - 0... = Don't fragment: Not set
 - 0... = More fragments: Not set
 - ...0 0000 0000 0000 = Fragment Offset: 0
 - Time to live: 64
 - Protocol: ICMP (1)
 - Header checksum: 0x03cf [validation disabled]
 - [Header checksum status: Unverified]
 - Source Address: 192.168.86.1
 - Destination Address: 192.168.86.61
 - Internet Control Message Protocol
 - Data (38 bytes)
- Packet Bytes:** Displays the raw data in hexadecimal and ASCII. The first few bytes are 78 4f 43 98 09 27 3c 20 64 00 0e c0 00 00 00 00, which correspond to the Ethernet II header fields.

```

> Frame 51: 98 bytes on wire (784 bits), 98 bytes captured (784 bits) on interface en0, id 0
> Ethernet II, Src: Google_89:be:c8 (3c:2d:89:be:c8), Dst: Apple_98:d9:27 (78:4f:43:98:d9:27)
> Internet Protocol Version 4, Src: 192.168.86.1, Dst: 192.168.86.61
    0100 .... = Version: 4
    > >>> 0101 = Header Length: 20 bytes (5)
    > Differentiated Services Field: 0x00 (DSCP: CS6, ECN: Not-ECT)
        Total Length: 84
        Identification: 0x6880 (26768)
    > 000. .... = Flags: 0x0
        0... .... = Reserved bits: Not set
        .0... .... = Don't fragment: Not set
        ..0... .... = More fragments: Not set
        ...0 0000 0000 0000 = Fragment Offset: 0
        Time to Live: 64
        Protocol: TCP (1)
        Header Checksum: 0x63e6 [validation disabled]
        [Header checksum status: Unverified]
        Source Address: 192.168.86.1
        Destination Address: 192.168.86.61
> Internet Control Message Protocol

```

ip-wireshark-trace1-1.pcapng

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ip.dst==192.168.86.61 and icmp

No.	Time	Source	Destination	Protocol	Length	Info
45	1.888088	192.168.86.1	192.168.86.61	ICMP	98	Time-to-live exceeded (Time to live exceeded in transit)
49	1.875315	192.168.86.1	192.168.86.61	ICMP	98	Time-to-live exceeded (Time to live exceeded in transit)
51	1.876637	192.168.86.1	192.168.86.61	ICMP	98	Time-to-live exceeded (Time to live exceeded in transit)
53	1.888044	192.168.86.1	192.168.86.61	ICMP	98	Time-to-live exceeded (Time to live exceeded in transit)
57	1.888090	192.168.86.1	192.168.86.61	ICMP	98	Time-to-live exceeded (Time to live exceeded in transit)
59	1.892580	192.168.86.1	192.168.86.61	ICMP	98	Time-to-live exceeded (Time to live exceeded in transit)
61	1.908367	192.168.86.9	192.168.86.61	ICMP	70	Time-to-live exceeded (Time to live exceeded in transit)
63	1.927998	192.168.86.9	192.168.86.61	ICMP	70	Time-to-live exceeded (Time to live exceeded in transit)
65	1.948119	192.168.86.9	192.168.86.61	ICMP	70	Time-to-live exceeded (Time to live exceeded in transit)
68	1.958959	68.87.181.185	192.168.86.61	ICMP	98	Time-to-live exceeded (Time to live exceeded in transit)
70	1.965187	68.87.181.185	192.168.86.61	ICMP	98	Time-to-live exceeded (Time to live exceeded in transit)
72	1.975638	68.87.181.185	192.168.86.61	ICMP	98	Time-to-live exceeded (Time to live exceeded in transit)
74	1.980744	68.110.23.181	192.168.86.61	ICMP	110	Time-to-live exceeded (Time to live exceeded in transit)
76	1.988760	68.110.23.181	192.168.86.61	ICMP	110	Time-to-live exceeded (Time to live exceeded in transit)
78	2.024870	68.110.23.181	192.168.86.61	ICMP	110	Time-to-live exceeded (Time to live exceeded in transit)
80	2.024952	192.168.86.61	192.168.86.61	ICMP	98	Time-to-live exceeded (Time to live exceeded in transit)

```
> Frame 53: 98 bytes on wire (784 bits), 98 bytes captured (784 bits) on Interface em0, id 0
> Ethernet II, Src: Google_B9:bc:b3 (3c:20:6d:b9:bc:b3), Dst: Apple_08:09:27 (78:4f:43:08:09:27)
> Internet Protocol Version 4, Src: 10.0.0.1, Dst: 192.168.0.61
    000 .... = Version: 4
    <<<< 0000 = Header Length: 20 bytes (5)
    > Differentiated Services Field: 0xc0 (DSCP: CS6, ECN: Not-ECT)
        Total Length: 64
        Identification: 0xd5c3 (54723)
    <<<< 000 .... = Flags: 0x0
        0 .... = Reserved bit: Not set
        .0. .... = Don't fragment: Not set
        ..0. .... = More fragments: Not set
        ...0 0000 0000 0000 = Fragment Offset: 0
        Time to Live: 63
        Protocol: ICMP (1)
        Header Checksum: 0xb43f [validation disabled]
        [Header checksum status: Unverified]
        Source Address: 10.0.0.1
        Destination Address: 192.168.0.61
    > Internet Control Message Protocol
    Data (28 bytes)
```

ip-wireshark-trace1-1.pcapng

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ip.dst==192.168.86.61 and icmp

No.	Time	Source	Destination	Protocol	Length	Info
45	1.068888	192.168.86.1	192.168.86.61	ICMP	98	Time-to-live exceeded (Time to live exceeded in transit)
49	1.073315	192.168.86.1	192.168.86.61	ICMP	98	Time-to-live exceeded (Time to live exceeded in transit)
51	1.076637	192.168.86.1	192.168.86.61	ICMP	98	Time-to-live exceeded (Time to live exceeded in transit)
53	1.080429	10.0.0.1	192.168.86.61	ICMP	98	Time-to-live exceeded (Time to live exceeded in transit)
57	1.080900	10.0.0.1	192.168.86.61	ICMP	98	Time-to-live exceeded (Time to live exceeded in transit)
59	1.092580	10.0.0.1	192.168.86.61	ICMP	98	Time-to-live exceeded (Time to live exceeded in transit)
61	1.906167	96.120.66.9	192.168.86.61	ICMP	70	Time-to-live exceeded (Time to live exceeded in transit)
63	1.927998	96.120.66.9	192.168.86.61	ICMP	70	Time-to-live exceeded (Time to live exceeded in transit)
66	1.948130	96.120.66.9	192.168.86.61	ICMP	70	Time-to-live exceeded (Time to live exceeded in transit)
68	1.950559	68.87.181.105	192.168.86.61	ICMP	98	Time-to-live exceeded (Time to live exceeded in transit)
70	1.965187	68.87.181.105	192.168.86.61	ICMP	98	Time-to-live exceeded (Time to live exceeded in transit)
72	1.975638	68.87.181.105	192.168.86.61	ICMP	98	Time-to-live exceeded (Time to live exceeded in transit)
74	1.990744	96.110.23.101	192.168.86.61	ICMP	110	Time-to-live exceeded (Time to live exceeded in transit)
76	2.008780	96.110.23.101	192.168.86.61	ICMP	110	Time-to-live exceeded (Time to live exceeded in transit)
78	2.024870	96.110.23.101	192.168.86.61	ICMP	110	Time-to-live exceeded (Time to live exceeded in transit)
80	2.044952	162.151.52.226	192.168.86.61	ICMP	98	Time-to-live exceeded (Time to live exceeded in transit)

> Frame 61: 70 bytes on wire (560 bits), 70 bytes captured (560 bits) on interface en0, id 0
Ethernet II, Src: Google_89:0e:c8 (3c:28:6d:89:0e:c8), Dst: Apple_9b:d9:27 (78:4f:43:9b:d9:27)
Internet Protocol Version 4, Src: 96.120.66.9, Dst: 192.168.86.61
0100 = Version: 4
.... 0101 = Header Length: 20 bytes (5)
> Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
Total Length: 56
Identification: 0x0000 (0)
0000 = Flags: 0x0
0... = Reserved bit: Not set
.0... = Don't Fragment: Not set
..0... = More Fragments: Not set
...0 0000 0000 0000 = Fragment Offset: 0
Time to Live: 253
Protocol: ICMP (1)
Header Checksum: 0x45e (validation disabled)
[Header checksum status: Unverified]
Source Address: 96.120.66.9
Destination Address: 192.168.86.61
Internet Control Message Protocol

0000 78 4f 43 9b d9 27 3c 28 6d 89 0e c8 00 00 00 00 xOCC'(\ m-----E-
0010 00 38 00 00 00 00 00 fd 01 04 5e 68 70 42 09 c0 a8 -Th-@-----V---
0020 56 3d 06 00 81 9f 00 00 00 00 45 00 00 38 fd a7 V-----E-B-
0030 00 00 01 11 2f a8 c0 a8 56 3d 00 77 f5 0c fd a0 ---f---Vw-----
0040 82 9d 00 24 f2 fd 00 00 00 00 00 00 00 00 00 00 --\$-----
0050 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
0060 00 00 ..

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ip.dst==192.168.86.61 and icmp

No.	Time	Source	Destination	Protocol	Length	Info
45	1.068888	192.168.86.1	192.168.86.61	ICMP	98	Time-to-live exceeded (Time to live exceeded in transit)
49	1.073315	192.168.86.1	192.168.86.61	ICMP	98	Time-to-live exceeded (Time to live exceeded in transit)
51	1.076637	192.168.86.1	192.168.86.61	ICMP	98	Time-to-live exceeded (Time to live exceeded in transit)
53	1.080429	10.0.0.1	192.168.86.61	ICMP	98	Time-to-live exceeded (Time to live exceeded in transit)
57	1.080900	10.0.0.1	192.168.86.61	ICMP	98	Time-to-live exceeded (Time to live exceeded in transit)
59	1.092580	10.0.0.1	192.168.86.61	ICMP	98	Time-to-live exceeded (Time to live exceeded in transit)
61	1.906167	96.120.66.9	192.168.86.61	ICMP	70	Time-to-live exceeded (Time to live exceeded in transit)
63	1.927998	96.120.66.9	192.168.86.61	ICMP	70	Time-to-live exceeded (Time to live exceeded in transit)
66	1.948130	96.120.66.9	192.168.86.61	ICMP	70	Time-to-live exceeded (Time to live exceeded in transit)
68	1.950559	68.87.181.105	192.168.86.61	ICMP	98	Time-to-live exceeded (Time to live exceeded in transit)
70	1.965187	68.87.181.105	192.168.86.61	ICMP	98	Time-to-live exceeded (Time to live exceeded in transit)
72	1.975638	68.87.181.105	192.168.86.61	ICMP	98	Time-to-live exceeded (Time to live exceeded in transit)
74	1.990744	96.110.23.101	192.168.86.61	ICMP	110	Time-to-live exceeded (Time to live exceeded in transit)
76	2.008780	96.110.23.101	192.168.86.61	ICMP	110	Time-to-live exceeded (Time to live exceeded in transit)
78	2.024870	96.110.23.101	192.168.86.61	ICMP	110	Time-to-live exceeded (Time to live exceeded in transit)
80	2.044952	162.151.52.226	192.168.86.61	ICMP	98	Time-to-live exceeded (Time to live exceeded in transit)
82	2.062965	162.151.52.226	192.168.86.61	ICMP	98	Time-to-live exceeded (Time to live exceeded in transit)
84	2.081232	162.151.52.226	192.168.86.61	ICMP	98	Time-to-live exceeded (Time to live exceeded in transit)
86	2.101239	96.100.47.146	192.168.86.61	ICMP	110	Time-to-live exceeded (Time to live exceeded in transit)
88	2.122132	96.100.47.146	192.168.86.61	ICMP	110	Time-to-live exceeded (Time to live exceeded in transit)
90	2.141117	96.100.47.146	192.168.86.61	ICMP	110	Time-to-live exceeded (Time to live exceeded in transit)
92	2.160917	96.222.38.42	192.168.86.61	ICMP	70	Time-to-live exceeded (Time to live exceeded in transit)
94	2.180722	96.222.38.42	192.168.86.61	ICMP	70	Time-to-live exceeded (Time to live exceeded in transit)
96	2.199197	96.222.38.42	192.168.86.61	ICMP	70	Time-to-live exceeded (Time to live exceeded in transit)
98	2.220121	192.60.83.113	192.168.86.61	ICMP	70	Time-to-live exceeded (Time to live exceeded in transit)
100	2.242064	192.60.83.113	192.168.86.61	ICMP	70	Time-to-live exceeded (Time to live exceeded in transit)

> Frame 51: 98 bytes on wire (784 bits), 98 bytes captured (784 bits) on interface en0, id 0
Ethernet II, Src: Google_89:0e:c8 (3c:28:6d:89:0e:c8), Dst: Apple_9b:d9:27 (78:4f:43:9b:d9:27)
Internet Protocol Version 4, Src: 192.168.86.1, Dst: 192.168.86.61
0100 = Version: 4
.... 0101 = Header Length: 20 bytes (5)
> Differentiated Services Field: 0xc0 (DSCP: CS6, ECN: Not-ECT)
Total Length: 84
Identification: 0x0880 (2673)
0000 = Flags: 0x0
...0 0000 0000 0000 = Fragment Offset: 0
Time to Live: 64
Protocol: ICMP (1)
Header Checksum: 0xe1ce (validation disabled)
[Header checksum status: Unverified]
Source Address: 192.168.86.1
Destination Address: 192.168.86.61
Internet Control Message Protocol
Data (28 bytes)

0000 78 4f 43 9b d9 27 3c 28 6d 89 0e c8 00 00 45 c0 xOCC'(\ m-----E-
0010 00 54 68 80 00 00 00 00 e3 ce c0 a8 56 01 c0 a8 -Th-@-----V---
0020 56 3d 06 00 81 9f 00 00 00 00 45 00 00 38 fd a3 V-----E-B-
0030 00 00 01 11 2f a8 c0 a8 56 3d 00 77 f5 0c fd a0 ---f---Vw-----
0040 82 9d 00 24 f2 fd 00 00 00 00 00 00 00 00 00 00 --\$-----
0050 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
0060 00 00 ..

Packets: 317 - Displayed: 72 (22.7%)

Profile: Default

1:56 PM 11/9/2023

Wireshark interface showing a packet capture of ICMP Echo (ping) requests from 192.168.86.61 to 192.168.86.61. The capture shows multiple packets, all of which are marked as "Time to live exceeded (Time to live exceeded in transit)".

The packet list shows the following details for the selected packet (No. 78):

- Frame 78: 110 bytes on wire (880 bits), 110 bytes captured (880 bits) on interface en0, id 0
- Ethernet II, Src: Google_89:de:c8 (3c:20:6d:89:de:c8), Dst: Apple_98:d9:27 (78:4f:43:98:d9:27)
- Internet Protocol Version 4, Src: 192.168.86.61, Dst: 192.168.86.61
- ICMP Echo (ping) request, Id: 0, Seq: 0, Unchecked Sum: 0
- Header Length: 20 bytes (5)
- Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
- Total Length: 96
- Identification: 0x2099 (11161)
- Flags: 0x0
- Reserved bit: Not set
- Don't fragment: Not set
- More fragments: Not set
- Fragment Offset: 0
- Time to Live: 251
- Protocol: ICMP (1)
- Header Checksum: 0x054b [validation disabled]
- Header checksum status: Unverified
- Source Address: 192.168.86.61
- Destination Address: 192.168.86.61
- Internet Control Message Protocol
- Data (28 bytes)

The packet details pane shows the raw data of the packet, including the Ethernet II header, Internet Protocol Version 4 header, and ICMP Echo (ping) request data.

Part 2: Fragmentation

The image shows a Wireshark packet capture of an IP fragmentation event. The packet list pane at the top shows packet 179, which is a fragmented IP packet (protocol=UDP, offset=0, ID=64929) [Reassembled in #181]. The packet details pane for packet 179 shows the IP header with the 'More Fragments' flag set. The packet bytes pane shows the raw data of the packet, which is a 1488-byte UDP datagram. The packet is fragmented into two parts: a first fragment of 1488 bytes (offset 0) and a second fragment of 1488 bytes (offset 1488). The packet is captured on interface en0, id 0.

13. Find the first IP datagram containing the first part of the segment sent to 128.119.245.12 sent by your computer via the `traceroute` command to `gaia.cs.umass.edu`, *after* you specified that the `traceroute` packet length should be 3000. (Hint: This is packet 179 in the `ip-wireshark-trace1-1.pcapng` trace file.). Has that segment been fragmented across more than one IP datagram?

ANSWER:

Yes, packet 179 segment has been fragmented across more than one IP datagram.

14. What information in the IP header indicates that this datagram been fragmented?

ANSWER:

More fragment is set which indicates that this datagram has been fragmented.

15. What information in the IP header for this packet indicates whether this is the first fragment versus a latter fragment?

ANSWER:

Fragment offset: 0 indicates that packet 179 is 1st packet.

16. How many bytes are there in this IP datagram (header plus payload)?

ANSWER:

Total length: 1500bytes (header(20 bytes)+payload(1480 bytes)).

The image shows a Wireshark packet capture of a fragmented UDP packet. The packet list shows two fragments: packet 179 (offset 0) and packet 180 (offset 1480). The packet details for packet 180 show the IP header, UDP header, and the fragmented data payload. The packet details for packet 179 show the IP header, UDP header, and the fragmented data payload. The packet details for packet 180 show the IP header, UDP header, and the fragmented data payload.

Packet 179: 1514 bytes on wire (12112 bits), 1514 bytes captured (12112 bits) on interface en0, id 0
Ethernet II, Src: Apple_90:8d:92:78, Dst: Google_08:00:0c:3c:20:3d, Len: 1500
Internet Protocol Version 4, Src: 192.168.86.61, Dst: 128.119.245.12
0000 = Version: 4
.... 0010 = Header Length: 20 bytes (5)
Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
Total Length: 1500
Identification: 0x6da2 (64930)
0001 = Flags: 0x0, More fragments
0... = Reserved bit: Not set
0... = Don't fragment: Not set
...1 = More fragments: Set
...0 0000 1001 1001 = Fragment Offset: 1480
Time to Live: 1
Protocol: UDP (17)
Header Checksum: 0x094c [validation disabled]
[Header checksum status: Unverified]
Source Address: 192.168.86.61
Destination Address: 128.119.245.12
[Reassembled IPv4 in frame #181]
Data (1480 bytes)

17. Now inspect the datagram containing the second fragment of the fragmented UDP segment. What information in the IP header indicates that this is *not* the first datagram fragment?

ANSWER:

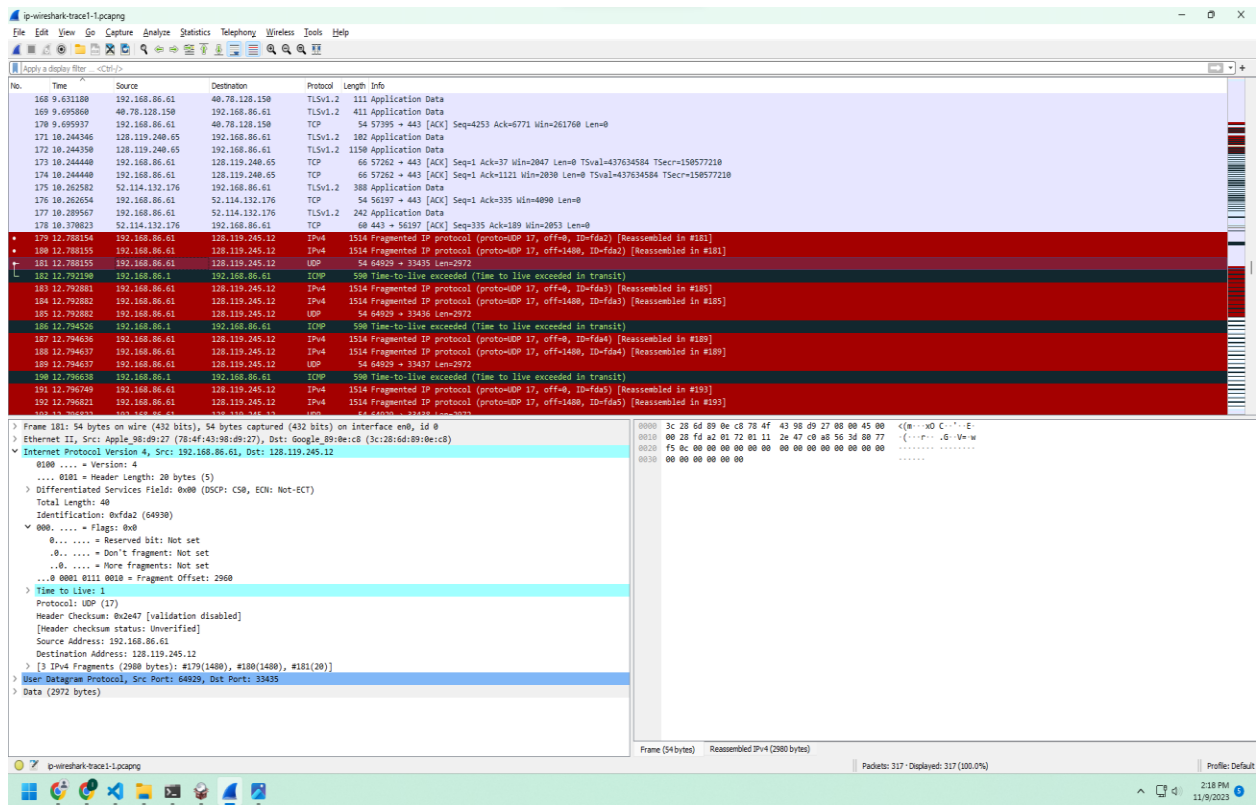
Fragment offset: 1480 indicates that this is not 1st fragment.

18. What fields change in the IP header between the first and second fragment? List ALL the fields that change in the IP header.

ANSWER:

Fields change in the IP header between the first and second fragment

- Fragment offset
- Header Checksum



19. Now find the IP datagram containing the third fragment of the original UDP segment.
 What information in the IP header indicates that this is the last fragment of that segment?
ANSWER:
 Fragment offset: 2960, so it is the 3rd fragment and More fragment is not set(0) indicates that this fragment is the last fragment.